Stage 1 Capstone — Submission Template

Applied Cryptography · Ciphers & Secrets

> **How to use this template.** Replace every `[ bracketed prompt ]` with your own work, then delete the grey instruction lines (the ones that begin with *Instructions —*) before you submit. You may hand this in as **one combined document** or split it into the four files **D1–D4** — graders accept either, as long as all four areas are covered. Every claim needs an evidence citation (the artefact file it came from) **and**, where research applies, one external standard cited by section number (RFC / NIST SP / FIPS). Made-up section numbers score zero.

Field	Your entry
Full name	[your full name]
Intern ID (UBI-2026-XXXX)	[your intern ID]
Stage	Stage 1 — Applied Cryptography
Date	[DD-MM-YYYY]

D1 — Crypto failure mapping

> **Instructions —** The post-mortem. Name **every** cryptographic failure in The Griot's artefacts, one row per finding. Each finding cites the specific artefact file it came from **and** one external standard by section number. Open with two or three sentences on hash vs encryption, using the Griot's own choices as the example.

Hash vs encryption (2–3 sentences): [When should the Griot have hashed, and when encrypted? Use their own choices.]

#	Finding	Evidence (artefact file)	External standard (section #)	Why it matters
1	[e.g. AES key + IV stored in the same staging config]	[01-aes-ciphertext.txt]	[e.g. NIST SP 800-57 §…]	[…]
2	[e.g. reused / predictable IV]	[…]	[…]	[…]
3	[e.g. HS256 signed with a 5-	[04-weak-jwt-hmac.txt]	[e.g. RFC 7518 §…]	[…]

#	Finding	Evidence (artefact file)	External standard (section #)	Why it matters
	character secret]			
4	[e.g. JWT alg=none / missing exp / privileged claim]	[03-jwts.txt]	[e.g. RFC 7519 §…]	[…]
5	[add rows as needed]	[…]	[…]	[…]

D2 — Decoded artefact appendix

> **Instructions —** The working. For each artefact give the recovered plaintext, the method you used, and the intermediate outputs where a file has layers. Show your working — plaintext alone is not enough.

Artefact 01 — AES-CBC ciphertext

- Method (CyberChef recipe URL or Python script): […]
- Recovered plaintext: […]

Artefact 02 — Classical cipher

- Cipher identified (Caesar / Vigenère / substitution): […]
- How you identified it (one sentence — frequency analysis is plenty): […]
- Recovered plaintext: […]

Artefact 03 — Three JWTs

Token	Header (alg)	Key payload claims	Red flag
A	[…]	[…]	[…]
B	[…]	[…]	[…]
C	[…]	[…]	[…]

Artefact 05 — Layered reconnaissance memo

- Layer 1 — base64 → [intermediate output]
- Layer 2 — ROT13 → [intermediate output]
- Layer 3 — substitution → [final plaintext]

> **Instructions —** If you also cracked the weak HS256 token (artefact 04) and forged a token, paste the

> recovered secret and the forged token here — it supports a finding in D1.

D3 — Five controls the board should approve

> **Instructions —** Five concrete controls (products, policies, or configurations). Each maps to **at least two** failures from D1 and is specific enough that someone else could go and build or buy it. Assume a small team and a tight budget — "improve crypto hygiene" scores zero.

#	Control (specific)	Maps to D1 findings (#)	One-sentence justification
1	[e.g. rotate to AES-GCM with a managed KMS key]	[#1, #2]	[…]
2	[…]	[…]	[…]
3	[…]	[…]	[…]
4	[…]	[…]	[…]
5	[…]	[…]	[…]

D4 — Ethics stance: the Ethnos Cyber call

> **Instructions —** 300–500 words, **your own writing** — this is the document graders scan hardest for AI patterns. A senior contact at Ethnos Cyber (your cohort's sponsor) asks you to forward your findings privately, ahead of the Sankofa board briefing. Cover: who you consult before deciding; what you will and will not share, and why; how you weigh loyalty to the sponsor against your obligations to Sankofa; and one action you take regardless of what your manager says. Cite at least one ISC2 Code of Ethics canon **by number**.

[Your stance — 300–500 words, in your own voice]

ISC2 canon cited: [Canon …]

Before you submit — quick checklist

- ☐ Header filled in (name + intern ID) on page 1
- ☐ Every D1 finding cites an artefact file **and** an external standard by section number
- ☐ D2 shows the method and the intermediate layers, not just the plaintext
- ☐ Each D3 control maps to at least two D1 findings
- ☐ D4 is 300–500 words, in your own voice, with an ISC2 canon cited by number
- ☐ All instruction lines and `[ bracketed prompts ]` deleted
- ☐ Folder (and every file inside it) shared as **anyone with the link can view**